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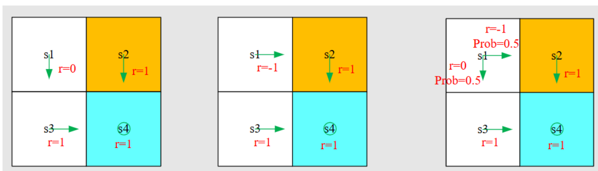
§ Lec II

Bellman Equation

1. Why is return important?

- Return could be used to evaluate policies.



Starting from s_1, the discounted returns are:

policy 1 (left figure)

$$\begin{aligned}\text{return}_1 &= 0+\gamma^1+\gamma^2+\cdots \\ &= \gamma(1+\gamma+\gamma^2+\cdots) \\ &= \frac{\gamma}{1-\gamma}\end{aligned}$$

policy 2 (middle figure)

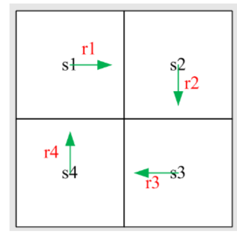
$$\begin{aligned}\text{return}_2 &= -1+\gamma^1+\gamma^2+\cdots \\ &= -1+\gamma(1+\gamma+\gamma^2+\cdots) \\ &= -1+\frac{\gamma}{1-\gamma}\end{aligned}$$

policy 3(stochastic) (right figure)

$$\begin{aligned}\text{return}_3 &= 0.5\left(-1+\frac{\gamma}{1-\gamma}\right)+0.5\left(\frac{\gamma}{1-\gamma}\right) \\ &= -0.5+\frac{\gamma}{1-\gamma}\end{aligned}$$

As $\text{return}_1 > \text{return}_2 > \text{return}_3$, the first policy is the best.

- How to calculate return?



- Method 1: by definition

Let v_i be the return value starting from s_i

$$\begin{aligned}v_1 &= r_1+\gamma r_2+\gamma^2 r_3+\cdots \\ v_2 &= r_2+\gamma r_3+\gamma^2 r_4+\cdots \\ v_3 &= r_3+\gamma r_4+\gamma^2 r_1+\cdots \\ v_4 &= r_4+\gamma r_1+\gamma^2 r_2+\cdots\end{aligned}$$

- Method 2: ***Bootstrapping*** !

The returns rely on each other.

$$\begin{aligned}v_1 &= r_1+\gamma v_2 \\ v_2 &= r_2+\gamma v_3 \\ v_3 &= r_3+\gamma v_4 \\ v_4 &= r_4+\gamma v_1\end{aligned}$$

How to solve the equations? Use matrices.

$$\underbrace{\begin{pmatrix}v_1\\v_2\\v_3\\v_4\end{pmatrix}}_{\text{v}} = \underbrace{\begin{pmatrix}r_1\\r_2\\r_3\\r_4\end{pmatrix}}_{\text{r}} + \begin{pmatrix}\gamma v_2\\\gamma v_3\\\gamma v_4\\\gamma v_1\end{pmatrix} = \underbrace{\begin{pmatrix}r_1\\r_2\\r_3\\r_4\end{pmatrix}}_{\text{r}} + \gamma \underbrace{\begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix}}_{\text{P}} \underbrace{\begin{pmatrix}v_1\\v_2\\v_3\\v_4\end{pmatrix}}_{\text{v}}$$

which can be rewritten as

$$\boxed{\boldsymbol{v}=\boldsymbol{r}+\gamma\boldsymbol{P}\boldsymbol{v}}$$

This is the **Bellman equation** (for this specific deterministic problem)!

It demonstrates the *core idea*: the value of one state *relies on* the values of other states.
A *matrix-vector* form is more clear to see how to solve the state values.

2. State Value

- Some Notations

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1}$$

S_t: state at time t
A_t: the action taken at state S_t
R_{t+1}: the reward obtained after taking A_t
S_{t+1}: the state transited to after taking A_t

Note that S_t, A_t, R_{t+1} are all *random variables*. (we can operate on them)

This single-step process is governed by the following probability distributions:

$S_t \to A_t$ is governed by $\pi(A_t = a \mid S_t = s)$ (policy)

$S_t, A_t \to R_{t+1}$ is governed by $p(R_{t+1} = r \mid S_t = s, A_t = a)$(reward prob)

$S_t, A_t \to S_{t+1}$ is governed by $p(S_{t+1} = s' \mid S_t = s, A_t = a)$(state transition prob)

Then, consider the following multi-step trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \ldots$$

The discounted return is

$$G_t = R_{\{t+1\}} + \gamma R_{\{t+2\}} + \gamma^2 R_{\{t+3\}} + \cdots$$

$\gamma \in [0, 1)$ is a discount rate.
G_t is also a *random variable* since $R_{\{t+1\}}, R_{\{t+2\}}, \cdots$ are random variables.

- The expectation of G_t is defined as the **state value**:

$$\boxed{\boldsymbol{v_{\pi(s)} = E[G_t \mid S_t = s]}}$$

- It is a function of s. It is a conditional expectation with the condition that ***the state starts from*** s.
- The state value is based on the policy π.
- If the state value is greater, then the policy is better because greater cumulative rewards can be obtained.

Q: What is the relationship between ***return*** and ***state value***?

A: The state value is *the mean of all possible returns* that can be obtained starting from a state. If everything — $\pi(a|s)$, $p(r|s,a)$, $p(s'|s,a)$ — is deterministic, then state value is the same as return.

3. Derivation of Bellman Equation

Consider a random trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \ldots$$

The return G_t can be written as

$$
\begin{aligned}
G_t &= R_{\{t+1\}} + \gamma R_{\{t+2\}} + \gamma^2 R_{\{t+3\}} + \cdots, \\
&= R_{\{t+1\}} + \gamma\left(R_{\{t+2\}} + \gamma R_{\{t+3\}} + \cdots\right), \\
&= R_{\{t+1\}} + \gamma G_{\{t+1\}},
\end{aligned}
$$

Then, it follows from the definition of the state value that

$$
\begin{aligned}
v_{\pi(s)} &= \mathbb{E}[G_t \mid S_t = s] \\
&= \mathbb{E}\left[R_{\{t+1\}} + \gamma G_{\{t+1\}} \mid S_t = s\right] \\
&= \mathbb{E}\left[R_{\{t+1\}} \mid S_t = s\right] + \gamma \mathbb{E}\left[G_{\{t+1\}} \mid S_t = s\right]
\end{aligned}
$$

Next, calculate the two terms, respectively.

① First, calculate $\mathbb{E}\left[R_{\{t+1\}} \mid S_t = s\right]$:

$$
\begin{aligned}
\mathbb{E}\left[R_{\{t+1\}} \mid S_t = s\right] &= \sum_a \pi(a|s) \mathbb{E}\left[R_{\{t+1\}} \mid S_t = s, A_t = a\right] \\
&= \sum_a \pi(a|s) \sum_r p(r|s,a) r
\end{aligned}
$$

Notice: This is the mean of ***immediate rewards***. (the reward at s)

② Second, calculate $\mathbb{E}\left[G_{\{t+1\}} \mid S_t = s\right]$:

$$
\begin{aligned}
\mathbb{E}\left[G_{\{t+1\}} \mid S_t = s\right] &= \sum_{s'} \mathbb{E}\left[G_{\{t+1\}} \mid S_t = s, S_{\{t+1\}} = s'\right] p(s' \mid s) \\
&= \sum_{s'} \mathbb{E}\left[G_{\{t+1\}} \mid S_{\{t+1\}} = s'\right] p(s' \mid s) \\
&= \sum_{s'} v_{\pi(s')} p(s' \mid s) \\
&= \sum_{s'} v_{\pi(s')} \sum_a p(s' \mid s, a) \pi(a|s)
\end{aligned}
$$

Notice:

Due to the *memoryless* Markov property:

$$
\mathbb{E}\left[G_{\{t+1\}} \mid S_t = s, S_{\{t+1\}} = s'\right] = \mathbb{E}\left[G_{\{t+1\}} \mid S_{\{t+1\}} = s'\right]
$$

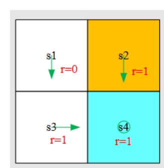
And the second term means the mean of ***future rewards***.

Therefore, we have the **Bellman Equation**:

$$
\begin{aligned}
v_{\pi(s)} &= \mathbb{E}\left[R_{\{t+1\}} | S_t = s\right] + \gamma \mathbb{E}\left[G_{\{t+1\}} | S_t = s\right], \\
&= \underbrace{\sum_a \pi(a|s) \sum_r p(r|s,a) r}_{\text{mean of immediate rewards}} + \gamma \underbrace{\sum_a \pi(a|s) \sum_{s'} p(s'|s,a) v_{\pi(s')}}_{\text{mean of future rewards}}, \\
&= \sum_a \pi(a|s) \left[\sum_r p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) v_{\pi(s')}\right], \forall s \in \mathcal{S}.
\end{aligned}
$$

- It characterizes the relationship among the *state-value functions of different states.* ($v_\pi(s)$ and $v_\pi(s')$)
- A set of equations: *every state* has an equation like this !!!

- Now we can calculate all state values using Bootstrapping !



- With the Bellman Equation, we get:

$$
\begin{aligned}
v_\pi(s_1) &= 0 + \gamma v_\pi(s_3), \\
v_\pi(s_2) &= 1 + \gamma v_\pi(s_4), \\
v_\pi(s_3) &= 1 + \gamma v_\pi(s_4), \\
v_\pi(s_4) &= 1 + \gamma v_\pi(s_4).
\end{aligned}
$$

- The equation can be solved as:

$$
\begin{aligned}
v_\pi(s_4) &= \frac{1}{1-\gamma}, \\
v_\pi(s_3) &= \frac{1}{1-\gamma}, \\
v_\pi(s_2) &= \frac{1}{1-\gamma}, \\
v_\pi(s_1) &= \frac{\gamma}{1-\gamma}.
\end{aligned}
$$

4. Matrix-Vector form for Bellman Equation

- How to solve the Bellman equation?

$$v_\pi(s) = \sum_a \pi(a|s)\left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a)v_\pi(s')\right]$$

① The above elementwise form is valid for every state $s \in \mathcal{S}$. That means there are $|\mathcal{S}|$ equations like this!
② If we put all the equations together, we have a set of linear equations, which can be concisely written in a matrix-vector form.

Rewrite the Bellman equation as

$$\boxed{v_\pi(s) = r_\pi(s) + \gamma \sum_{s'} p_\pi(s' \mid s)v_\pi(s')}$$

where

$$r_\pi(s) \triangleq \sum_a \pi(a \mid s) \sum_r p(r \mid s, a)r$$
$$p_\pi(s' \mid s) \triangleq \sum_a \pi(a \mid s)p(s' \mid s, a)$$

Suppose the states could be indexed as s_i ($i = 1, ..., n$).
For state s_i, the Bellman equation is

$$\boxed{v_\pi(s_i) = r_\pi(s_i) + \gamma \sum_{s_j} p_\pi(s_j \mid s_i)v_\pi(s_j)}$$

Put all these equations for all the states together and rewrite to a matrix-vector form

$$\boxed{\vec{v_\pi} = \vec{r_\pi} + \gamma P_\pi \vec{v_\pi}}$$

where

$\vec{v_\pi} = [v_\pi(s_1), ..., v_\pi(s_n)]^\top \in \mathbb{R}^n$

$\vec{r_\pi} = [r_\pi(s_1), ..., r_\pi(s_n)]^\top \in \mathbb{R}^n$

$P_\pi \in \mathbb{R}^{n \times n}$, where $[P_\pi]_{ij} = p_\pi(s_j \mid s_i)$, is the state transition matrix

If there are four states, $v_\pi = r_\pi + \gamma P_\pi v_\pi$ can be written out as

$$\underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi} = \underbrace{\begin{bmatrix} r_\pi(s_1) \\ r_\pi(s_2) \\ r_\pi(s_3) \\ r_\pi(s_4) \end{bmatrix}}_{r_\pi} + \gamma \underbrace{\begin{bmatrix} p_\pi(s_1|s_1) & p_\pi(s_2|s_1) & p_\pi(s_3|s_1) & p_\pi(s_4|s_1) \\ p_\pi(s_1|s_2) & p_\pi(s_2|s_2) & p_\pi(s_3|s_2) & p_\pi(s_4|s_2) \\ p_\pi(s_1|s_3) & p_\pi(s_2|s_3) & p_\pi(s_3|s_3) & p_\pi(s_4|s_3) \\ p_\pi(s_1|s_4) & p_\pi(s_2|s_4) & p_\pi(s_3|s_4) & p_\pi(s_4|s_4) \end{bmatrix}}_{P_\pi} \underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi}.$$

- For the matrix-vector form:

$$\vec{v_\pi} = \vec{r_\pi} + \gamma P_\pi \vec{v_\pi}$$

  - The closed-form solution is:

$$\boxed{\vec{v_\pi} = (I - \gamma P_\pi)^{-1} \vec{r_\pi}}$$

In practice, we still need to use numerical tools to calculate the matrix inverse.

  - By iterative algorithms, we can avoid the matrix inverse.

An iterative solution is:

$$\overrightarrow{v_{k+1}} = \vec{r_\pi} + \gamma P_\pi \vec{v_k}$$

Initially, we can input a simple v_0, and do the iteration.

This leads to a sequence

$$\{v_0, v_1, v_2, ...\}$$

. We can show that

$$v_k \to v_\pi = (I - \gamma P_\pi)^{-1} r_\pi, \quad k \to \infty$$

Iterating like this, v_k converges to our previous closed-form solution (provable).

5. Action Value

- ***State value***: the average return the agent can get starting from a state.
- ***Action value***: the average return the agent can get starting from a state and taking an action.

Why do we care action value?
Because we want to know which action is better.

Definition:

$$\boxed{q_\pi(s,a) = \mathbb{E}[G_t \mid S_t = s, A_t = a]}$$

- $q_\pi(s,a)$ is a function of the state-action pair (s,a) and it also depends on π(policy).

It follows from the properties of conditional expectation that

$$\underbrace{\mathbb{E}[G_t \mid S_t = s]}_{v_\pi(s)} = \sum_a \underbrace{\mathbb{E}[G_t \mid S_t = s, A_t = a]}_{q_\pi(s,a)} \pi(a|s)$$

Hence,

$$\boxed{v_\pi(s) = \sum_a \pi(a|s) q_\pi(s,a) \qquad (1)}$$

(state value can be divided by different actions)

Recall that the state value is given by

$$v_\pi(s) = \sum_a \pi(a|s) \underbrace{\left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s')\right]}_{q_\pi(s,a)}$$

By comparing them, we have the action-value function as

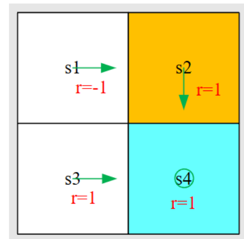
$$q_\pi(s,a) = \sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \quad (2)$$

Actually, the two expressions (1)&(2) shows two sides of the same coin:

① (1)shows how to obtain state values from action values.(this is simple because we just do average on the actions)

② (2)shows how to obtain action values from state values.(this one is harder)

- An illustrative example



With

$$q_\pi(s,a) = \sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s')$$

we can get:

$$q_\pi(s_1,a_2) = -1 + \gamma v_\pi(s_2)$$

Notice: although the policy given asks to take action a_2 at s_1, we can get action values for *other actions* !!!

$$\begin{aligned} q_\pi(s_1,a_1) &= -1 + \gamma v_\pi(s_1), \\ q_\pi(s_1,a_3) &= 0 + \gamma v_\pi(s_3), \\ q_\pi(s_1,a_4) &= -1 + \gamma v_\pi(s_1), \\ q_\pi(s_1,a_5) &= 0 + \gamma v_\pi(s_1). \end{aligned}$$

① We can first calculate all the state values and then calculate the action values.

② And we can also directly calculate the action values with or without models !

Q: Why can we compute $q_\pi(s, a)$ for an action a that π(the policy) never takes at state s?

A: $q_\pi(s, a)$ conditions on $A_t = a$ only for the current step. After that, all future actions follow π.

- It does not require $\pi(a|s) > 0$.
- It answers: "What if we deviate ***now***, then follow π?"
- This allows comparing actions to improve π.

Hence, evaluating actions not taken by π is both valid and essential.